

Finite Relational Selection Theory - FRST v1.0

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****Status:**** independent candidate mathematical ontology frozen before UCD comparison.

Central proposition

****Reality is modeled as one finite, relationally constrained quantum system.**** Space, time, particles, fields, geometry, classical records and probability are treated as organizations or observables of that system rather than separate primitive substances.

The fundamental hierarchy is

mathematical possibility $\rightarrow$ physical admissibility $\rightarrow$ actual universal state/history $\rightarrow$ observable reality.

Equivalently,

$\Omega_{\text{realized}} \subseteq \Omega_{\text{physical}} \subseteq \Omega_{\text{mathematical}}$.

Twelve frozen axioms

1. ****Finite actuality.**** There exists a finite set V of primitive realized identities: $0 < |V| = N < \text{infinity}$. Each primitive identity has a finite-dimensional local state space.
2. ****Relational primacy.**** Primitive labels have no physical meaning apart from invariant relations.
3. ****Completed carrier.**** Once the physical carrier exists, later evolution does not manufacture additional primitive identities.
4. ****Finite relational structure.**** A finite relational complex $K=(V,E,F,...)$ carries possible pairwise and higher-order relations.
5. ****Constraint selection.**** Not every mathematically writable configuration is physically admissible. Physical states satisfy a family of constraints $C_{\alpha} |\Psi\rangle = 0$.
6. ****Complex state principle.**** Coherent alternatives are represented in a complex Hilbert space.
7. ****Closed universal evolution.**** The complete system evolves within its physical Hilbert space without a fundamental external environment or external collapse law.
8. ****Locality of law.**** Fundamental interactions have bounded relational support or decay sufficiently rapidly with relational distance.
9. ****Internal time.**** Physical time is reconstructed from subsystem change/correlation rather than assumed as an external primitive clock.
10. ****Relational geometry.**** Spatial geometry is reconstructed from relational/link state rather than assumed as a pre-existing container.
11. ****Probability consistency.**** Probabilities are normalized, additive on mutually exclusive physical alternatives and consistent under equivalent representations of the same projector/event.
12. ****Observable closure.**** Every physically meaningful quantity belongs to the invariant physical observable algebra.

Raw mathematical possibility space

Give each primitive identity v a finite local state space $H_v \simeq C^{(d_v)}$ and each allowed relation e a finite relation state space $H_e \simeq C^{(r_e)}$. The unconstrained state space is

$$H_{\text{raw}} = (\text{tensor}_{\{v \text{ in } V\}} H_v) \text{ tensor } (\text{tensor}_{\{e \text{ in } E\}} H_e) \text{ tensor } \dots$$

Its dimension is finite:

$$D_{\text{raw}} = \text{product}_v d_v \text{ product}_e r_e \dots < \text{infinity}.$$

A normalized vector or density operator on H_{raw} is mathematically writable. That does not mean it is physically realizable.

Physical selector

Let the physical consistency constraints be C_{α} . Define

$$H_{\text{phys}} = \text{intersection}_{\alpha} \ker(C_{\alpha})$$

and let P_{phys} be the orthogonal projector onto that intersection. Then

$$P_{\text{phys}}^2 = P_{\text{phys}}, \quad P_{\text{phys}}^{\dagger} = P_{\text{phys}},$$

and a state is physically admissible exactly when

$$P_{\text{phys}} |\Psi\rangle = |\Psi\rangle.$$

This implements the distinction

$$H_{\text{raw}} \rightarrow [P_{\text{phys}}] H_{\text{phys}}.$$

Observable algebra and gauge redundancy

The full mathematical operator algebra is $B(H_{\text{raw}})$. The constrained physical observable algebra is schematically

$$A_{\text{phys}} = \{ O = P_{\text{phys}} O P_{\text{phys}} : U_g O U_g^{\dagger} = O \}.$$

Local-frame transformations are interpreted as representation redundancies when all physical observables are invariant under them.

If a relational transporter compares local frames at identities i and j , it transforms as

$$U_{ij} \rightarrow g_i U_{ij} g_j^{-1}.$$

For a closed relational path γ ,

$$W_{\gamma} = U_{i1i2} U_{i2i3} \dots U_{ini1}$$

transforms by conjugation, so $\text{Tr}(W_{\gamma})$ is invariant. Curvature can therefore be represented as nontrivial closed relational transport rather than assumed as a fundamental continuum gauge field.

Dynamics

The physical Hamiltonian must preserve the physical subspace:

$$[H, P_{\text{phys}}] = 0.$$

A relationally local form is

$$H = P_{\text{phys}} \left(\sum_{\{X \subset K\}} h_X \right) P_{\text{phys}}.$$

The universal state evolves as

$$\rho(s) = \exp(-isH) \rho_0 \exp(isH).$$

The parameter s orders the mathematical evolution; a physical clock does not have to be primitive.

Internal time

Split a subsystem notionally into clock C and remainder R . A globally constrained/stationary state can encode correlations such that the conditional state of R when the clock reads τ obeys an effective Schrodinger evolution. Thus

ordered change + clock correlation $\rightarrow$ physical time.

Time is an internal relational observable/order parameter rather than necessarily a substance in which the universe sits.

Relational causality

A local finite-range or sufficiently decaying Hamiltonian bounds the propagation of influence. Distant operators remain approximately commuting outside an effective relational causal cone. Schematically,

$$|[A_X(t), B_Y]| \leq C \exp[-\mu(d(X, Y) - v|t|)].$$

Effective light-cone behavior can therefore emerge from relational locality.

Relational geometry

Let every physical relation/edge have a positive length observable L_e . Define

$$\ell_e(\rho) = \text{Tr}(\rho L_e).$$

The distance between identities is the minimum weighted relational path:

$$d_{\rho}(i, j) = \min_{\{\gamma: i \rightarrow j\}} \sum_{\{e \in \gamma\}} \ell_e(\rho).$$

The primitive carrier may stay fixed while the relational metric changes. Therefore cosmic expansion does not require continued creation of primitive identities.

Effective dimension

Let Δ_{ρ} be a weighted graph/relational Laplacian and

$$P(\sigma) = (1/N) \text{Tr} \exp(-\sigma \Delta_{\rho}).$$

Define the spectral dimension

$$d_s(\sigma) = -2 \, d \ln P / d \ln \sigma.$$

An extended regime with $d_s \approx 4$ would behave effectively as four-dimensional geometry without assuming that dimension as a primitive input.

Fields and continuum

A continuum field is interpreted as a coarse collective observable of many relational degrees of freedom. For a region R ,

$$\Phi_R = (1/|R|) \sum_{v \in R} O_v.$$

A successful continuum approximation should admit an error estimate such as

$$|O_{\text{discrete}}(L) - O_{\text{continuum}}(L)| \leq C (\ell/L)^p$$

for $L \gg \ell$. The continuum is therefore an effective mathematical envelope, not automatically an ontology of infinitely many physical points.

Matter, charge and mass

Stable localized excitation sectors of the physical Hamiltonian represent matter. Conserved symmetry-sector labels represent charges. If

$$H|\phi_n\rangle = E_n|\phi_n\rangle$$

and E_0 is the ground energy, then the first excitation cost is

$$\Delta = E_1 - E_0.$$

At an effective relativistic level one may identify

$$m c_{\text{eff}}^2 = \Delta.$$

Mass is therefore represented as a spectral cost of leaving the ground relational sector rather than as an independent primitive substance.

Finite spectral-gap example

Suppose

$$H = \Delta_0 \sum_v Q_v + \sum_e J_e + \sum_f K_f,$$

with all local terms positive semidefinite and annihilating the common ground sector, and suppose every normalized state orthogonal to the ground sector activates at least one Q_v . Then

$$\langle \psi | H | \psi \rangle \geq \Delta_0$$

for every normalized excited state, so

$$\text{gap}(H) \geq \Delta_0 > 0.$$

This is a finite-system theorem template; no physical infinite-volume ontology is required to state it.

Measurement, probability and classicality

Measurement is modeled as unitary correlation of system, apparatus and environment. Environment-induced decoherence can suppress interference between macroscopically distinct record sectors while the full universal state remains lawful.

Under the standard finite-dimensional quantum probability premises, the probability of projector P is

$$p(P) = \text{Tr}(\rho P).$$

For a pure state and one-dimensional outcome projector this yields the Born form $|c_a|^2$.

Classical facts are stable, redundantly recorded relational information: an approximately commuting, dynamically robust record algebra that can be copied into multiple disjoint environmental fragments.

Thermodynamics and the arrow of time

Global von Neumann entropy is constant under closed unitary evolution, while subsystem/coarse entropy can increase as correlations and entanglement spread. Macroscopic irreversibility is therefore effective and depends on special low-correlation boundary conditions rather than fundamental destruction of information.

Gravity and singularities

Matter and relational geometry are degrees of freedom of the same global system. A successful low-energy limit must reproduce the observed gravitational dynamics with controlled corrections. Literal continuum singularities are not fundamental objects in a finite carrier with finite-dimensional local state spaces and bounded physical generators; they indicate breakdown of the continuum approximation.

Completed creation versus history

Completed creation fixes the primitive carrier $K=(V,E,F,\dots)$. History changes the state ρ . Thus

$$K(t) = K_C$$

can hold after completion while

$$\rho(t_1) \neq \rho(t_2).$$

Finished creation is therefore mathematically compatible with continuing evolution, relational expansion, structure formation, interaction and decay.

Master FRST object

The theory is summarized by

$$R_{\text{FRST}} = (K, H_{\text{raw}}, P_{\text{phys}}, A_{\text{phys}}, H, \rho_0, \text{observables})$$

with finite primitive carrier throughout.

Observable physics is a map from this one object into effective descriptions of time, space, geometry, fields, particles, mass, charge, probability, records and classical history.

Frozen boundary

FRST v1.0 is a mathematical architecture, not an experimentally established theory of our universe. It does not yet derive the numerical carrier, local dimensions, exact physical constraint family, exact Hamiltonian/couplings, or the actual initial/boundary state.

Its deepest unresolved specification problem is why one particular triple

$(P_{\text{phys}}, H, \rho_0)$

or an equivalent unified law is selected rather than another.

Most importantly for the comparison project, **FRST v1.0 assumes finite actuality. UCD must not import that assumption as a theorem.** The UCD VS FRST program instead asks whether Unified Coherence Dynamics can derive finite realization from its own source law and whether that same source law can force the specific Q72 -> H504 -> H-net architecture.